

Chanuga Liyanage

Mechanical Engineering Undergraduate Specializing in Mechatronics

✉ chanugaliyanagepro@gmail.com | 📞 +94 70 487 2742 | 🌐 LinkedIn | 🐙 GitHub | 🌐 Portfolio

96/A, Galle Road, Pohaddaramulla, Wadduwa, Kalutara

EDUCATION

University of Peradeniya

2022 – 2026

B.Sc. Engineering (Hons) in Mechanical Engineering, Specialized in Mechatronics

GPA: 3.60 / 4.00

EXPERIENCE

Vega Innovations

July - October 2024

Battery System Engineering Intern

- Gained hands-on experience in electric vehicle (EV) technology and mechanical systems, enhancing understanding of how EVs and industrial machines function.
- Contributed to various stages of design and development, supported technical operations, and deepened knowledge through both engineering and project management skills.

KSPA Panadura

September - November 2025

Mechanical Engineering Intern

- Gained hands-on experience in corrugated manufacturing processes and industrial machinery, developing understanding of machine mechanisms, production workflows, and system operations.
- Observed machine maintenance and troubleshooting processes, and supported technical analysis of cutting tools and automation components used in manufacturing, strengthening knowledge of mechanical systems and industrial operations.

PROJECTS

RideTwin: AI-Based Digital Twin for Vehicle Ride Comfort Analysis

Final Year Research

- Developed a physics-based digital twin of a quarter-car suspension using MATLAB/Simulink, applied ISO 2631 ride comfort evaluation and signal processing, and built machine learning models for ride comfort prediction and suspension parameter optimization.

Hybrid Electric Vehicle (HEV) Controller Design

Group Project

- Designed and implemented a microcontroller-based control system for a series hybrid electric vehicle prototype, validated battery and motor interaction through MATLAB/Simulink simulations, and developed a scaled-down working model demonstrating real-time power source switching and speed control.

Cell Compression Machine Design

Intern

- Designed a pneumatic cell compression machine to apply constant and adjustable compression force to LiFePO₄ battery cells, improving their cycle life.

V6 Engine Modeling and Assembly using SolidWorks

Individual

- Modeled and assembled a functional 3D V6 engine in SolidWorks, gaining a comprehensive understanding of the design and operation of individual engine components.

AWARDS

- First Place at the Skyward Challenge organized by KDU IESL at the Techno 2024 Exhibition.
- First Place at Gravity Games 2024 organized by SEDS Earth in collaboration with SEDS Sri Lanka.
- First Runners-up at the Aero Day Quiz Competition organized by Hybrid Education in collaboration with RC Falcons.

TRAINING & CERTIFICATIONS

Industrial Automation Workshop (PLC and SCADA)

July 2024

- Conducted by Eng. Sugath Pradeep, Head of the Technical Industry Business, IET University of Peradeniya. Gained practical knowledge of PLC and SCADA technologies through interactive demonstrations and real-world industrial applications.

Supervised Machine Learning: Regression and Classification

June 2024

DeepLearning.AI & Stanford University (Coursera)

- Developed supervised machine learning models in Python using NumPy and Scikit-learn for regression and binary classification tasks, including linear and logistic regression.

Advanced Learning Algorithms

June 2024

DeepLearning.AI & Stanford University (Coursera)

- Built and trained neural network models using TensorFlow for multi-class classification and applied machine learning best practices to improve model generalization and real-world performance.

VOLUNTEERING

- Committee Member, Automotive Society, University of Peradeniya *Present*
- Member, Students for the Exploration and Development of Space (SEDS), University of Peradeniya *Present*
- Member, AeroGenZ Society, University of Peradeniya *Present*

EXTRACURRICULAR ACTIVITIES

Sports

- **Weightlifting**
 - 2nd Place, Inter-Faculty Freshers' Weightlifting Championship 2023

REFERENCES

Prof. Primal Fernando

PhD Specialization in Thermodynamics and Refrigeration

Head of the Department

Department of Mechanical Engineering

University of Peradeniya, Sri Lanka

wpd@pdn.ac.lk

Phone: +94 71 845 0438

Prof. Asanga Ratnaweera

B.Sc. Eng. (Peradeniya), PhD (Melbourne)

Department of Mechanical Engineering

University of Peradeniya, Sri Lanka

asanga@eng.pdn.ac.lk